

Scalability & Future Plan

Date	12 June 2026
Team ID	SWTID-2026-9325
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	SQLite database is suitable for small-scale applications but not for large numbers of concurrent users.	Performance may decrease as the number of users and transactions grows.	High
2	The application currently supports only borrower and advisor roles with limited financial services.	Limits the system's usability for banks and financial institutions.	Medium
3	The application is designed for local deployment and lacks cloud hosting and monitoring.	Reduced scalability, availability, and accessibility for large-scale usage.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users.	Deploy using cloud platforms (AWS, Azure, or Google Cloud) with load balancing to support more users.
2	Data Storage	Uses SQLite database.	Migrate to PostgreSQL or MySQL for improved scalability, concurrency, and reliability.
3	Performance	Runs on a single FastAPI server.	Optimize API performance using caching, asynchronous processing, and containerization with Docker.
4	Security	Uses JWT authentication and role-based access control.	Add multi-factor authentication (MFA), HTTPS, audit logging, and encryption for enhanced security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Email/SMS notifications, payment reminders, and lender communication integration.	Next 3–6 Months	Improve borrower engagement and timely repayment.
Phase 3	Cloud deployment, PostgreSQL database migration, and mobile application support.	Next 6–12 Months	Increase scalability, accessibility, and overall system performance.
Phase 4	Advanced AI analytics, credit score prediction, multilingual support, and banking/payment gateway integration.	Beyond 12 Months	Provide intelligent financial insights and support a broader range of users and financial institutions.